WHAT IS LUMPY SKIN DISEASE?

- It is an extremely contagious disease caused by a virus that belongs to the same family as the smallpox and monkeypox viruses
- It spreads rapidly among cows and buffaloes through vectors like houseflies, mosquitoes and ticks
- It causes blister-like nodules all over the body, gradually opening up like deep wounds
- Symptoms include fever and excessive nasal and salivary secretion, followed by secondary infections in animals with low immunity
- There's no standard cure; a few allopathic and ayurvedic medicines have shown positive results
- Just like in Covid, sequestering the infected cattle and mass vaccination hold the key to containing the virus spread

ORIGIN AND OUTBREAKS

- LSD first originated in Africa, where the first epidemic was recorded in Zambia in 1929
- By 1989, it had covered the entire continent before venturing further, into Israel
- In recent years, cases have surfaced across West Asia and Europe
- In India, LSD was first reported in 2019, believed to have entered via Bangladesh. It moved towards the central and southern regions, covering 15 states by 2021
- In 2022, the first case surfaced in Gujarat in April. It is believed to have its source in Pakistan
- Since July, the outbreak has spread to Rajasthan, Punjab, Haryana, UP, Uttarakhand, MP and J&K, with stray cases reported from a few other states too

to the same family (poxviridae) as the smallpox and monkeypox viruses, LSD spreads among cattle through vectors such as houseflies, mosquitoes and ticks (see *What is Lumpy Skin Disease?*). Native to Africa, LSD was first reported in India in 2019 and had spread to 15 states by 2021, but mortality was never so high. In 2022, Gujarat reported the first case in April, though the alarm bells started ringing when the disease further spread to Rajasthan and at least nine other states/ UTs in July-August (see *Origin and Outbreaks*).

The outbreak is particularly severe in Rajasthan. "Lumpy came like a wildfire, consuming our livestock," rues Meenakshi Meena, the sarpanch of Rajawas, an affluent village on the outskirts of Jaipur, as she tends to the open wounds of her cattle. In just under two months, the viral disease has infected 1.3 million of the 14 million cattle in the state, leaving about 60,000 dead. The official toll across India—with deaths reported in at least seven other states—has crossed 70,000, though the actual figures, say experts, could be much higher.

Even Prime Minister Narendra Modi acknowledged the severity of the outbreak, when he addressed the World Dairy Summit in Greater Noida in Uttar Pradesh on September 12: "In the recent past, there has been loss of livestock in many states of India due to the disease named lumpy. The central government, along with various state governments, is trying to control it. Our scientists have also developed a vaccine for it."

The vaccine that the PM was talking of is Lumpi-ProVacInd, indigenously developed by two institutes under the Indian Council of Agricultural Research. Unlike in other countries, which resort to culling of infected bovines to curb the virus spread, mass vaccination and sequestering the cattle remains the only option in India. Since the commercial launch of the Lumpi-ProVacInd vaccine could take several months, India has launched a massive drive meanwhile using the goat pox vaccine that, according to experts, is just "60 to 70 per cent effective" against LSD.

In Rajasthan, orders have been placed for 4.1 million vaccines with a million already administered, says Lal Chand Kataria, the state's minister for animal husbandry. However, with carcasses rotting in the open and multiple mutations of the virus fuelling fears of a fresh wave, the Rajasthan High Court on September 19 told the state government to ensure that the uninfected bovines are "adequately" vaccinated and the dead ones are buried safely.

THE DETECTION OF AT LEAST SIX MUTATIONS OF THE VIRUS IN RAJASTHAN HAS FUELLED FEARS OF A FRESH WAVE

DOUBLE WHAMMY

In Rajasthan, the wave of cattle deaths originated along its western border with Pakistan and surged through various districts, moving in crests and troughs, before hitting Jaipur with full intensity earlier this month. The outbreak has left the state's rural economy devastated, since, according to government data, 80 per cent of the rural households in Rajasthan keep livestock, and the animal husbandry sector alone contributes to about 10 per cent of the state's gross domestic product. Even as the state accounts for only 7 per cent of the country's cattle population, it contributes to nearly 13 per cent of India's total milk production, second only to Uttar Pradesh.

Rural households, therefore, have been hit with a double whammy, says Dr N.M. Singh, additional director, animal husbandry, Rajasthan. "The death of a domesticated cow often leads to a lot of trauma in the owner's family because of both the emotional attachment as well as the loss of investment and steady income," he says. About